Practice problems #17

1. Consider the same matrix that was given below.

$$A = \begin{bmatrix} 1 & 3 & 5 \\ -1 & -3 & 1 \\ 0 & 2 & 3 \\ 1 & 5 & 2 \\ 1 & 5 & 8 \end{bmatrix}$$

   (a) Explain why the given matrix has a QR factorization.

   (b) Find an upper triangular matrix R with positive diagonal entries so that $A = QR$.

2. Not every application needs the diagonals of the matrix R to be positive. If we drop this requirement, you don't need to update Q, and there is an easy way of computing the matrix R. What is an easy way? Discuss.

Answers or Hints on page 2.

#1(a): Columns of A are?

#1(b): Find R using the formula for r_{ij} from the class, and check if your R is correct by computing QR (it should equal A).

#2: Since $A = QR$, we have $Q^T A = Q^T QR$.
Note that the matrix Q has _____columns, so $Q^T Q$ =?